MIL-E-1/1504B
9 March 1981

SUPERSEDING
MIL-E-1/1504A
10 November 1970

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5965

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Twin triode, miniature, computer service

Outline	---	6-2(EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	ehk	Ik/k	Pp/a	Ic/g	Rg/g	Alt
Unit:	V	Vdc	Vdc	Vdc	v	mAdc	W	mAdc	Meg	ft
Maximum:	13.2	330	0	+100	+200	18.0	2.2	0.5	0.1	(see
	6.6	---	---	---	---	---	(see note 1)	---	(see note 2)	note 3)
Minimum:	12.0	---	-75	-200	-200	---	---	---	---	---
	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3 (see note 4)	150	-2	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Rg = 0.1 Meg (see notes 5 and 6)	0.65	II	Ic	0	-1.0	µAdc
1261	Electrode voltage (grid)	Eb = 150 Vdc; Ec/Ib = 150 µA (see note 5)	0.65	II	Ec	---	-7.5	Vdc
1301	Heater current		0.65	II	If	415	485	mAdc
1256	Electrode current (1) (anode)	See note 5	0.65	II	Ib	4.3	12.0	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes	See note 5	2.5	I	R	50	---	Meg
1336	Heater-cathode leakage	See note 5	2.5	I	Ihk	---	20	µAdc
1256	Electrode current (2) (anode)	Eb = 100 Vdc (see notes 5 and 7)	2.5	I	Ib	13.6	22.0	mAdc
1306	Transconductance (1)	See note 5	2.5	I	Sm	5,000	9,000	µmhos
1306	Transconductance (2)	Ef = 5.7 V (see note 5)	---	---	Sm	4,000	---	µmhos
1266	Grid emission	Ef = 7.0 V; Eb = 150 Vdc; Ec = -20 Vdc; Rg = 0.1 Meg (see notes 5 and 8)	---	---	Isg	---	-2.0	µAdc
1261	Electrode voltage (grid) (difference between sections)	Eb = 150 Vdc; Ec/Ib = 150 µAdc (see note 9)	---	---	Ec	---	2.0	v
1031	Low-frequency vibration	Rp = 2,000 ohms (see note 10)	---	---	Ep	---	300	mVac
2126	Envelope strain		---	---	---	---	---	---
1121	Base strain		---	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
1506	Heater-cycling life	Ef = 7.5 V; Ehk = +135 Vdc; Ec = Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:					
1336	Heater-cathode leakage		Ihk	---	60	µAdc

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued					
---	Life-test (1)	Ebb = 150 Vdc; Ec = 0; Rk/k = 39 ohms; Ehk = +100 Vdc; Rg/g = 0.1 Meg; Rb/a = 1,600 ohms (see notes 11 and 12)	---	---	---	---
---	Life-test (1) end points (1,000 hours):					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	---	-1.5	µAdc
1256	Electrode current (1) (anode)		Ib	4.0	---	mAdc
1256	Electrode current (2) (anode)		Ib	12.3	---	mAdc
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1211	Insulation of electrodes		R	10	---	Meg
1266	Grid emission (see note 13)		Isg	---	-4.0	µAdc
1256	Average change in electrode current (1) (anode) (see note 14)		Avg ΔIb	---	25	%
---	Total defectives		---	---	---	---
---	Life-test (2) (cutoff operation)	Eb = 150 Vdc; Ec = -75 Vdc; Ehk = -100 Vdc; Rg/g = 0.075 Meg; Rk/k = 15,000 ohms (see notes 11 and 12)	---	---	---	---
---	Life-test (2) end points (cutoff operation) (1,000 hours):					
---	Inoperatives		---	---	---	---
1266	Total grid current		Ic	---	-1.5	µAdc
1256	Electrode current (1) (anode)		Ib	4.0	---	mAdc
1256	Electrode current (2) (anode)		Ib	12.3	---	mAdc
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1211	Insulation of electrodes		R	10	---	Meg
1266	Grid emission (see note 13)		Isg	---	-4.0	µAdc
---	Total defectives		---	---	---	---
1511	Cathode interface life (see notes 5, 12, and 15)	Ef = 6.9 Vdc; Eb = 100 Vdc; Ec/Ib = 2.5 mAdc	Ri	---	25	Ohms

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NOTES:

1. The absolute-maximum rating for the combined anode dissipation of both anodes is 4.2 watts.
2. This value is for operation under fixed bias conditions. With cathode bias, Rg/g may be 0.5 Meg maximum.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
4. Tie heaters in parallel.
5. Test each section separately with elements of opposite section grounded.
6. This test shall be performed at the conclusion of the holding period.
7. Tie each grid through a 500,000-ohm resistor (±1 percent) to the corresponding anode.
8. Prior to this test, tubes shall be preheated a minimum of 5 minutes with all sections operating at the conditions specified below. The 3-minute test is not permitted. Test within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ecc Vdc	Rk/k Ohms	Rg/g Meg	Eb Vdc
7.0	0	150	0.5	150

9. Adjust bias on each section for Ib = 150 µA per section. Difference in bias between sections shall not exceed the limit specified.
10. Read with sections tied in parallel.
11. Operate each section separately with specified Rg in each grid and specified Rk in each cathode.
12. The life-test sample shall consist of 20 tubes for each life-test (1) and (2) selected at random in such a way that each tube in the lot has an equal chance of being selected. In the event of failure of the first sample, a second sample of 40 tubes shall be selected and subjected to the life-test in which the failure occurred. Acceptance shall then be based on the combined results from the first and second samples.
13. This test shall be performed on the life-test samples after all other specified tests at the initial and 1,000-hour reading periods.
14. The change in the average value of electrode current (1) (anode) characteristic of the total number of samples on life at zero to 1,000 hours shall be less than 25 percent down from the average value at zero hours.
15. Cathode interface resistance shall be read after short and discontinuity detection test and before all tests specified on the life-test (2) cutoff-operation life-test samples.
16. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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